Aziz Ben Ammar

Computational Biology PhD Candidate | Systems Biology · Metabolic Modeling · Scientific Python
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Skills

Programming Languages: Python, R, MATLAB
Data Science & Machine Learning: Pandas, NumPy, SciPy, scikit-learn, PyTorch, Matplotlib, Seaborn; data preprocessing, statistical analysis, regression, supervised learning, model evaluation.
Systems & Computational Biology: COBRApy, genome-scale metabolic modeling (GEM), flux balance analysis (FBA), flux variability analysis (FVA), enzyme-constrained modeling, protein allocation modeling, strain design
Workflow & Development Tools: Git, GitHub/GitLab, Snakemake, Jupyter, VS Code, Conda, Linux/HPC environments

Experience

Research Assistant, RWTH Aachen University – Aachen, Germany — January 2024 – Present

- Developed and validated a genome-scale metabolic model (GEM) of a bacterial production host using Python and COBRApy, achieving 95% predictive accuracy for experimental growth phenotypes and enabling in silico analysis of metabolic capabilities. (GitHub: iABA974).
- Designed and developed a reusable Python package for metabolic model curation and analysis, automating the retrieval and integration of biochemical information from BiGG and KEGG APIs to streamline model development and reduce manual curation (GitHub: afaa_gem).
- Applied model-driven strain design and genetic-algorithm optimization to identify metabolic engineering interventions for recombinant protein production, achieving a 30% increase in predicted product yield compared with the baseline strain.
- Extended the P. putida iJN1436 metabolic model for plastic upcycling by integrating PET-, PU-, and PBAT-derived monomer assimilation pathways and applied constraint-based simulations to identify optimal mixed-substrate compositions for PHB production (GitHub: iJN1463_plastic).
- Collaborated in a cross-functional computational–experimental team, translating model predictions into experimentally testable hypotheses and using experimental results to iteratively refine model design and predictions.
- Mentored six students through successful completion of their research projects and delivered university-level teaching, demonstrating technical leadership, knowledge transfer, and the ability to communicate complex quantitative concepts to different audiences.

Research Intern, Technical University of Dresden – Dresden, Germany — Feb 2023 – Sep 2023

- Built a reproducible NGS analysis workflow integrating quality control, genome assembly, functional annotation, and taxonomic profiling, with Snakemake for workflow automation and QIIME 2 for microbiome data analysis (GitHub: AutoQIIME2).
- Developed a Python/Tkinter desktop application for structured biogas metadata acquisition and downstream statistical analysis, enabling standardized data collection and more efficient processing of experimental datasets.

Publications

- **Ben Ammar**, A., et al. Genome-scale metabolic modeling of Xanthobacter sp. SoF1. mSystems (ASM). Manuscript under review.
- Bernal-Cabas, **M., Ben Ammar**, A., Terpstra, O., et al. (2025). Food production from air: gas precision fermentation with hydrogen-oxidising bacteria. Trends in Biotechnology, 44, 906–920.
- Meng, H., Karmainski, T., **Ben Ammar, A**., Sieberichs, A., et al. (2026). Engineering Pseudomonas putida KT2440 for open-loop upcycling of mixed plastics. bioRxiv, preprint 2026.03.23.713816.

Education

RWTH Aachen University – PhD in Computational Biology — Expected January 2027
Ecole Polytechnique de Sousse – Master in Biotechnology Engineering — September 2023

Languages

English (Fluent) · German (B2) · French (Native) · Arabic (Native)